

Topic 8 – Energy – forces doing work

Students should:	Maths skills
8.1 Describe the changes involved in the way energy is stored when systems change	
8.2 Draw and interpret diagrams to represent energy transfers	1c, 2c
8.3 Explain that where there are energy transfers in a closed system there is no net change to the total energy in that system	
8.4 Identify the different ways that the energy of a system can be changed a through work done by forces b in electrical equipment c in heating	
8.5 Describe how to measure the work done by a force and understand that energy transferred (joule, J) is equal to work done (joule, J)	
8.6 Recall and use the equation: work done (joule, J) = force (newton, N) × distance moved in the direction of the force (metre, m) $E = F \times d$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d 4f
8.7 Describe and calculate the changes in energy involved when a system is changed by work done by forces	
8.8 Recall and use the equation to calculate the change in gravitational PE when an object is raised above the ground: change in gravitational potential energy (joule, J) = mass (kilogram, kg) × gravitational field strength (newton per kilogram, N/kg) × change in vertical height (metre, m) $\Delta GPE = m \times g \times \Delta h$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
8.9 Recall and use the equation to calculate the amounts of energy associated with a moving object: kinetic energy (joule, J) = $\frac{1}{2} \times \text{mass (kilogram, kg)} \times (\text{speed})^2$ ((metre/second) ² , (m/s) ²) $KE = \frac{1}{2} \times m \times v^2$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
8.10 Explain, using examples, how in all system changes energy is dissipated so that it is stored in less useful ways	

Students should:	Maths skills
8.11 Explain that mechanical processes become wasteful when they cause a rise in temperature so dissipating energy in heating the surroundings	
8.12 Define power as the rate at which energy is transferred and use examples to explain this definition	1c
8.13 Recall and use the equation: power (watt, W) = work done (joule, J) ÷ time taken (second, s) $P = \frac{E}{t}$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
8.14 Recall that one watt is equal to one joule per second, J/s	1c
8.15 Recall and use the equation: efficiency = $\frac{\text{(useful energy transferred by the device)}}{\text{(total energy supplied to the device)}}$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d

Use of mathematics

- Make calculations of the energy changes associated with changes in a system, recalling or selecting the relevant equations for mechanical, electrical, and thermal processes; thereby express in quantitative form and on a common scale the overall redistribution of energy in the system (1a, 1c, 3c).
- Calculate relevant values of stored energy and energy transfers; convert between newton-metres and joules (1c, 3c).
- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).

Suggested practicals

- Investigate power by moving up the stairs, step-ups onto a low platform or lifting objects of different weights.